CAFFEINE & CADAVER
MBBS BIOCHEMISTRY NOTES SERIES

Molecular Biology & Genetics

MBBS BIOCHEMISTRY · PAPER 1

DNA Replication · Transcription · Translation · Lac Operon · PCR · DNA Repair · Mutations · Genetic Code · Antibiotic Targets · Post-translational Modification · Recombinant DNA

01 DNA Replication

Replication is **semi-conservative** (proven by Meselson-Stahl experiment), **bidirectional**, and proceeds **5'→3'** only. Template is read 3'→5'.

PROKARYOTIC (E. COLI) REPLICATION — KEY PLAYERS ⭐ PYQ 2012-S, 2019, 2021

PROTEIN/ENZYME	ROLE
DnaA protein	Recognises and binds origin (oriC)
Helicase (DnaB)	Unwinds double helix by breaking H-bonds — uses ATP
SSB proteins	Single-Strand Binding proteins — stabilise unwound strands, prevent re-annealing
Primase (DnaG)	Synthesises RNA primer (8–12 nucleotides) — needed because DNA pol cannot initiate de novo
DNA Pol III MAIN	Primary replicase — fast, processive, 5'→3' synthesis + 3'→5' proofreading exonuclease
DNA Pol I	Removes RNA primers (5'→3' exonuclease) + fills gap with DNA
DNA Pol II	DNA repair only
DNA Ligase	Seals nicks between Okazaki fragments — uses NAD^+ (prokaryotes) or ATP (eukaryotes)
Topoisomerase I	Relaxes supercoiling — cuts 1 strand, rotates, reseals (no ATP)
Topoisomerase II (Gyrase)	Relieves positive supercoiling ahead of fork — cuts both strands — ATP-dependent. Target of **fluoroquinolones** (prokaryotes) and **etoposide** (eukaryotes/anticancer)

▶ PROKARYOTIC REPLICATION — STEP BY STEP

Step 1 — Initiation **oriC** recognised by DnaA → Helicase loads (DnaB) → Replication bubble forms → Primase adds RNA primers on both strands

↓ Both strands serve as templates

Step 2 — Elongation **Leading strand:** continuous 5'→3' synthesis (DNA Pol III) **Lagging strand:** discontinuous — synthesised as **Okazaki fragments** (1000–2000 nt in prokaryotes, 100–200 in eukaryotes), each requires a new primer

↓ DNA Pol I removes primers, DNA Ligase seals nicks

Step 3 — Termination Two forks meet at **Ter sequences** (termination sites) → Tus protein arrests helicase → Topoisomerase II decatenates interlocked circles

PROKARYOTIC VS EUKARYOTIC REPLICATION PYQ 2015, 2021

FEATURE	PROKARYOTIC	EUKARYOTIC
Origin	Single (oriC)	Multiple origins (thousands)
Speed	~1000 nt/sec	~50–100 nt/sec (but multiple origins = faster overall)
Main polymerase	DNA Pol III	DNA Pol δ (lagging), DNA Pol ε (leading)
Primer removal	DNA Pol I (5'→3' exonuclease)	RNase H + FEN1
Okazaki fragments	1000–2000 nt	100–200 nt
Histones	Absent	Present — must reassemble after fork
Telomere problem	Circular DNA — no end problem	Linear — ends shorten → **Telomerase** (RNA-dependent DNA pol) extends them
Ligase cofactor	NAD^+	ATP

TELOMERASE & CANCER

Normal somatic cells have no telomerase → chromosomes shorten with each division → senescence. **Cancer cells reactivate telomerase** → maintain telomere length → become immortal. Stem cells also express telomerase. Target for anticancer therapy.

02 Transcription

Transcription = DNA → RNA. Reads template (antisense) strand **3'→5'**, synthesises mRNA **5'→3'**. No primer needed. Uses **ribonucleotides**. Only a small region of DNA is copied at a time.

PROKARYOTIC TRANSCRIPTION

PYQ 2010, 2013, 2024

RNA Polymerase (single enzyme does all):

- ✦ Core enzyme: $\alpha_2\beta\beta'\omega$
- ✦ σ (sigma) factor added → holoenzyme
- ✦ σ factor recognises **promoter** (–10 and –35 boxes)
- ✦ σ factor dissociates after initiation begins

Phases:

- ✦ **Initiation:** Holoenzyme binds promoter → DNA melts (~17 bp) → first phosphodiester bond forms
- ✦ **Elongation:** σ released → core enzyme elongates 5'→3'
- ✦ **Termination:** Rho-independent (hairpin + U-run) or Rho-dependent (Rho helicase)

EUKARYOTIC RNA POLYMERASES

PYQ 2017-S, 2024

POLYMERASE	LOCATION	PRODUCT	INHIBITOR
RNA Pol I	Nucleolus	28S, 18S, 5.8S rRNA	Actinomycin D (also Pol II)
RNA Pol II	Nucleoplasm	mRNA (hnRNA) + snRNA	**α-Amanitin** (most sensitive)
RNA Pol III	Nucleoplasm	tRNA, 5S rRNA, snRNA	α-Amanitin (high concentration)

MNEMONIC — RNA POL PRODUCTS

"I make Ribosomal, II makes Messenger, III makes Transfer" (1=rRNA, 2=mRNA, 3=tRNA)

EUKARYOTIC MRNA PROCESSING (POST-TRANSCRIPTIONAL)

HIGH YIELD

▶ HNRNA → MATURE MRNA

Step 1 — 5' Capping (co-transcriptional) 7-methylguanosine cap added to 5' end via 5'–5' triphosphate bridge
Protects from exonucleases; aids ribosome binding; required for translation

↓

Step 2 — 3' Polyadenylation Cleavage ~20 nt after AAUAAA signal → Poly-A polymerase adds 200–250 A residues
Stabilises mRNA; aids nuclear export; facilitates translation

↓

Step 3 — RNA Splicing (Intron removal) Spliceosome (snRNPs: U1,U2,U4,U5,U6) removes introns via 2-step transesterification
Lariat intermediate formed. Exons join. Alternative splicing → protein diversity

↓ Export to cytoplasm via nuclear pore

Mature mRNA: 5'cap – 5'UTR – Coding sequence – 3'UTR – Poly-A tail – 3'

RNA EDITING

ApoB mRNA editing: In intestine, C→U editing at codon 2153 of ApoB mRNA → Stop codon → **ApoB-48** (shorter) instead of ApoB-100. ApoB-48 is on chylomicrons; ApoB-100 on VLDL/LDL from liver. Same gene, tissue-specific protein — no alternate splicing.

03 Genetic Code & tRNA

PROPERTIES OF THE GENETIC CODE PYQ 2019-S

PROPERTY	MEANING	EXCEPTION/NOTE
Triplet	3 bases = 1 codon → 64 codons for 20 amino acids	No exceptions
Degenerate/Redundant	Multiple codons for same amino acid	3rd base wobble. Trp and Met have only 1 codon each
Non-overlapping	Each base belongs to only one codon	Some viruses have overlapping reading frames (exception)
Comma-free	No "punctuation" between codons, read continuously	Frameshift mutations disrupt this
Universal	Same code in all organisms	Minor exceptions: mitochondrial code (AUA=Met, UGA=Trp)
Non-ambiguous	One codon = only one amino acid	No exceptions
Start codon	AUG (Met; fMet in prokaryotes)	Rarely GUG (Val) in prokaryotes
Stop codons	UAA ("Ochre"), UAG ("Amber"), UGA ("Opal/Umber")	UGA can encode selenocysteine (21st AA)

STOP CODONS MNEMONIC

"UAA = U Are Away, UAG = U Are Gone, UGA = U Go Away"

Or: **UAA (Ochre), UAG (Amber), UGA (Opal)** — "OAO" or think "U-A-A is the usual stop"

TRNA STRUCTURE PYQ 2011, 2018-S

Cloverleaf Structure — 4 Arms:

✦ **Acceptor arm (3' CCA):** Amino acid attached to 3'-OH of terminal adenosine. All tRNAs end in CCA-3' (added post-transcriptionally)

✦ **Anticodon arm:** Contains the 3-base anticodon that pairs with mRNA codon

✦ **D-arm (DHU arm):** Contains dihydrouridine. Recognition site for aminoacyl-tRNA synthetase

✦ **TΨC arm:** Contains ribothymidine & pseudouridine. Ribosome binding

✦ **Variable loop:** Between anticodon and TΨC arms

Key Facts:

✦ Smallest RNA (~73–93 nucleotides)

✦ L-shaped 3D structure (cloverleaf folded)

✦ ~15% of total cellular RNA

✦ **Aminoacyl-tRNA synthetase** (one per amino acid) charges tRNA — uses ATP (→ AMP + 2Pi)

✦ Wobble position = 3rd base of anticodon (5' end of anticodon) — can pair with multiple bases

✦ Initiator tRNA in prokaryotes: **fMet-tRNA^{fMet}** (formyl-methionine)

04 Translation

Translation = mRNA → Protein. Occurs at ribosomes. Direction: N-terminus to C-terminus. Reads mRNA **5'→3'** .

RIBOSOME STRUCTURE

FEATURE	PROKARYOTIC	EUKARYOTIC
Overall	70S	80S
Small subunit	30S (16S rRNA + 21 proteins)	40S (18S rRNA + 33 proteins)
Large subunit	50S (23S + 5S rRNA + 31 proteins)	60S (28S + 5.8S + 5S rRNA + 49 proteins)
A site	Aminoacyl-tRNA binding	Same
P site	Peptidyl-tRNA (growing chain)	Same
E site	Exit — empty tRNA leaves	Same

MNEMONIC – RIBOSOME SITES

"APE" = A(minoacyl) → P(eptidyl) → E(xit)

▶ PROKARYOTIC TRANSLATION – 3 PHASES

INITIATION — Uses GTP + Initiation Factors (IF1, IF2, IF3) 30S + mRNA (Shine-Dalgarno sequence binds 16S rRNA) + fMet-tRNA → 30S initiation complex
+ 50S subunit → **70S initiation complex**
fMet-tRNA enters P site directly (unique to initiation)

↓ Elongation factors EF-Tu (GTP, brings aa-tRNA to A site), EF-Ts (regenerates), EF-G (GTP, translocation)

ELONGATION — 3 steps per cycle, each costs GTP **1. Decoding:** EF-Tu·GTP delivers aminoacyl-tRNA to A site → codon-anticodon check
2. Peptide bond formation: Peptidyl transferase (23S rRNA — ribozyme!) transfers growing chain from P-tRNA to A-tRNA
3. Translocation: EF-G·GTP moves ribosome 1 codon (3'→5' on mRNA) → A→P→E shift

↓ Until stop codon (UAA/UAG/UGA) enters A site

TERMINATION — Release Factors (RF1 recognises UAA/UAG; RF2 recognises UAA/UGA; RF3 is GTPase) RF binds A site → peptidyl transferase catalyses hydrolysis → polypeptide released
Ribosome dissociates: RRF (ribosome recycling factor) + EF-G + IF3

EUKARYOTIC VS PROKARYOTIC TRANSLATION DIFFERENCES

FEATURE	PROKARYOTIC	EUKARYOTIC
Coupling	Transcription + Translation coupled (polysome)	Separated in space/time (nucleus + cytoplasm)
Initiator tRNA	fMet-tRNA (formyl-methionine)	Met-tRNA (NOT formylated)
mRNA recognition	Shine-Dalgarno sequence + 30S	5' cap + Kozak sequence + 40S (cap-dependent scanning)
Initiation factors	IF1, IF2, IF3 (3)	eIF1, eIF2, eIF3, eIF4, eIF5 (many)
Polycistronic mRNA	Yes (one mRNA codes multiple proteins)	Usually monocistronic
Elongation factors	EF-Tu, EF-Ts, EF-G	eEF1α (=EF-Tu), eEF1βγ (=EF-Ts), eEF2 (=EF-G)
GTP used	2 per elongation cycle + 1 initiation	Similar

05 Lac Operon Regulation

Lac operon = inducible operon (in E. coli). Regulates genes for lactose metabolism. Has **negative** (repressor) AND **positive** (CAP/cAMP) regulation.

STRUCTURE OF THE LAC OPERON

▶ LAC OPERON ARCHITECTURE

CAP sitePositive control → Promoter (P)RNA Pol binds → Operator (O)Repressor binds →
lacZβ-galactosidase → lacYPermease → lacATransacetylase

4 STATES OF LAC OPERON EXPRESSION ⭐ PYQ 2015, 2018, 2019-S, 2023

CONDITION	REPRESSOR	CAP-CAMP	TRANSCRIPTION	LOGIC
No lactose + Glucose	Active (binds operator)	Absent (cAMP low)	**OFF**	No substrate needed + glucose preferred
Lactose + Glucose	Inactive (allolactose removes it)	Absent (cAMP still low)	**LOW**	No repressor BUT no positive activator either
No lactose + No glucose	Active (blocks operator)	Present (cAMP high, CAP binds)	**OFF**	CAP ready but repressor blocks — no point
Lactose + No glucose	Inactive (allolactose)	Present (cAMP high)	MAX ON	Repressor gone + CAP activates — cell uses lactose as energy source

Negative Regulation (Repressor):

✦ lacI gene encodes **lac repressor** (tetramer)

✦ Without lactose: repressor binds operator → blocks RNA Pol → no transcription

✦ Inducer = **allolactose** (isomer of lactose, formed by β-galactosidase acting on small amounts of lactose)

✦ Allolactose binds repressor → conformational change → repressor released from operator → transcription proceeds

✦ IPTG is a gratuitous inducer (non-metabolisable — used in experiments)

Positive Regulation (CAP-cAMP):

✦ Glucose present → cAMP LOW (adenylate cyclase inhibited) → CAP cannot bind → low transcription

✦ Glucose absent → cAMP HIGH → cAMP binds CAP (catabolite activator protein = CRP) → CAP-cAMP complex binds CAP site upstream of promoter

✦ CAP bends DNA → RNA Pol binds promoter more efficiently → **transcription stimulated ~50-fold**

✦ This is the basis of **catabolite repression** — glucose represses other sugar operons

LAC OPERON MEMORY HOOK

"No glucose = HIGH cAMP = CAP ON = Operon ready. Lactose present = Repressor OFF = GO. Both needed for MAX transcription."

06 DNA Damage & Repair

TYPES OF DNA DAMAGE PYQ 2011, 2017, 2023

TYPE	CAUSE	LESION
Deamination	Spontaneous (most common)	Cytosine → Uracil; Adenine → Hypoxanthine; 5-methylC → Thymine (G:C hotspot mutations)
Depurination	Spontaneous, acid	Loss of purine base — most common spontaneous lesion (~10,000/cell/day)
UV radiation	Sunlight (UV-B, 260 nm)	Pyrimidine dimers (TT > TC > CC), 6-4 photoproducts — adjacent thymines covalently linked
Alkylation	Alkylating agents (EMS, MMS, nitrogen mustard)	O^6-methylguanine (mispairs with T) → G:C → A:T transition
Oxidative damage	ROS (OH•)	8-oxoguanine (pairs with A), strand breaks
Intercalation	Acridine dyes, ethidium bromide	Base insertions → frameshift mutations
Ionising radiation	X-rays, γ-rays	Double-strand breaks (most lethal), base damage
Cross-linking	Cisplatin, mitomycin C	Intrastrand or interstrand cross-links

DNA REPAIR MECHANISMS ⭐ PYQ MULTIPLE YEARS

MECHANISM	LESION REPAIRED	KEY ENZYMES	DISEASE IF DEFECTIVE
Direct Reversal	O^6-methylguanine, UV dimers (prokaryotes)	O^6-methylG methyltransferase; Photolyase (uses visible light)	—
Base Excision Repair (BER)	Small, non-helix-distorting lesions (uracil, 8-oxoG, abasic sites)	DNA Glycosylase (removes base) → AP Endonuclease → DNA Pol β → DNA Ligase	MUTYH-associated polyposis (colon)
Nucleotide Excision Repair (NER)	Bulky, helix-distorting lesions (UV dimers, chemical adducts)	UvrABC excinuclease (prokaryotes); XP proteins in eukaryotes; excises ~12–13 nt (pro) or 24–29 nt (euk)	**Xeroderma Pigmentosum** (XPA–XPG mutations)
Mismatch Repair (MMR)	Replication errors: base mismatches, insertion/deletion loops	MutS (recognises), MutL (recruits), MutH (cuts unmethylated/new strand); MSH2, MLH1 in eukaryotes	**Lynch Syndrome** (HNPCC) — hereditary colorectal cancer
Double-Strand Break Repair — NHEJ	Double-strand breaks (DSBs)	Ku proteins + DNA-PKcs + ligase IV/XRCC4; error-prone	—
Homologous Recombination (HR)	DSBs in S/G2 phase (template available)	RAD51, BRCA1/BRCA2; error-free	BRCA1/2 mutations → breast/ovarian cancer
Translesion Synthesis	Replication past unrepaired lesions (SOS response)	Error-prone Pol IV, Pol V (prokaryotes); Rev1, Pol η, Pol ι (eukaryotes)	Pol η mutations → Xeroderma Pigmentosum variant

XERODERMA PIGMENTOSUM — NER DEFECT

Defect in **NER** → UV-induced pyrimidine dimers not removed → mutations accumulate in skin cells → extreme photosensitivity, freckling, skin cancers (melanoma, SCC, BCC) at very young age. Neurological deterioration in some types. 8 complementation groups (XPA–XPG + XPV).

MISMATCH REPAIR VS BASE EXCISION REPAIR — COMPARISON

PYQ 2019-S, 2023

FEATURE	MISMATCH REPAIR (MMR)	BASE EXCISION REPAIR (BER)
Lesion type	Replication mismatches, small indels	Damaged/abnormal bases (uracil, 8-oxoG)
Strand discrimination	Parental (methylated) vs new (unmethylated) in prokaryotes	N/A — single base removed from damaged strand
Key initiation	MutS recognises mismatch → MutL recruits MutH	Glycosylase recognises and removes damaged base → AP site
Excision length	Long tract (hundreds to thousands of nt)	Short patch (1 nt) or long patch (2–10 nt)
Key gap-filling polymerase	DNA Pol III (pro) / Pol δ (euk)	DNA Pol β (short patch) / Pol δ,ε (long patch)
Disease	Lynch syndrome (MMR gene mutations)	MUTYH colonic polyposis

07 Mutations

TYPE	DEFINITION	EFFECT ON PROTEIN	CLASSIC EXAMPLE
Silent/Synonymous	Base change → same amino acid (degenerate code)	No change	Most 3rd position changes
Missense	Base change → different amino acid	Altered protein (may be functional or non-functional)	HbS (Glu→Val), HbM
Nonsense	Base change → stop codon	Premature termination — truncated non-functional protein	β-Thalassaemia (some forms)
Frameshift	Insertion or deletion of 1 or 2 bases — shifts reading frame	Completely altered protein from mutation onwards; usually ends in premature stop	Creutzfeldt-Jakob-like if in prion gene; Duchenne MD (exon deletions)
Transition	Purine↔Purine (A↔G) or Pyrimidine↔Pyrimidine (C↔T)	Often missense	Most common point mutation type
Transversion	Purine↔Pyrimidine	Variable	Less common than transition
Trinucleotide repeat expansion	CAG repeat expansion	Polyglutamine tract → toxic gain-of-function	Huntington's (CAG), Fragile X (CGG), Myotonic dystrophy (CTG)

SICKLE CELL — POINT MUTATION (MISSENSE)

GAG → GTG (codon 6, β-globin). A→T transversion at 2nd position of codon. Glu (charged) → Val (hydrophobic). Results in HbS polymerisation when deoxygenated.

EQ: "Sickle cell anaemia is a molecular disease [2023]" — first disease understood at molecular level (Linus Pauling, 1949).

FRAMESHIFT VS POINT MUTATION IMPACT

Point mutation: changes 1 amino acid (may retain some function — e.g., missense). Frameshift: disrupts ALL downstream amino acids → completely non-functional truncated protein → usually more severe than point mutation.

Philadelphia chromosome [PYQ 2023]: t(9;22) translocation → BCR-ABL fusion gene → constitutively active tyrosine kinase → CML. Imatinib (Gleevec) specifically inhibits BCR-ABL kinase. Example of chromosomal translocation → oncogene activation.

08 PCR & Recombinant DNA

POLYMERASE CHAIN REACTION (PCR)

⭐ PYQ 2011-S, 2012-S, 2015-S, 2021

Components Required:

✦ Template DNA
✦ Two primers (forward + reverse, flanking target)
✦ **Taq polymerase** (thermostable from Thermus aquaticus) or Pfu
✦ dNTPs (dATP, dGTP, dCTP, dTTP)
✦ $MgCl_2$ (cofactor for Taq)
✦ Thermocycler

Applications:

✦ Diagnosis of infectious diseases (TB — gold standard [PYQ 2024])
✦ Prenatal genetic diagnosis
✦ DNA fingerprinting (forensics, paternity)
✦ Detection of mutations/SNPs
✦ Gene cloning / recombinant DNA
✦ Research: gene expression (RT-PCR)
✦ Cancer: detect BCR-ABL, HER2 amplification

▸ PCR CYCLE — 3 STEPS (REPEATED 25-35 CYCLES → 2^n COPIES)

Step 1 — Denaturation Heat to **94–96°C** → H-bonds broken → double helix separates into single strands

↓ Rapid cooling

Step 2 — Annealing Cool to **50–65°C** (depends on primer Tm) → primers bind complementary sequences on each strand

↓ Warm up

Step 3 — Extension Heat to **72°C** (Taq optimum) → Taq extends from 3' end of primer → new DNA strand synthesised 5'→3'

Note: Taq lacks 3'→5' proofreading (error-prone). For high-fidelity, Pfu polymerase is used.

RECOMBINANT DNA TECHNOLOGY

PYQ 2010-S, 2017-S

TOOL	FUNCTION	KEY DETAIL
Restriction Endonucleases	Cut DNA at specific palindromic sequences (4–8 bp)	Generate sticky ends (cohesive) or blunt ends. EcoRI: GAATTC↓CTTAAG. Named: organism-strain-type (e.g., EcoRI = E. coli R strain, first enzyme)
DNA Ligase	Joins cut DNA fragments	Seals phosphodiester bonds; requires ATP
Vectors	Carry foreign DNA into host	Plasmids (small), λ phage (medium), cosmids, YAC (yeast artificial chromosome — largest capacity [PYQ 2022])
Southern Blotting	Detect specific DNA sequence	DNA → gel electrophoresis → transfer to membrane → probe hybridisation [PYQ 2022, 2024]
Northern Blotting	Detect specific RNA	RNA → gel → transfer → probe
Western Blotting	Detect specific protein	Protein → SDS-PAGE → transfer → antibody probe
RFLP	Restriction Fragment Length Polymorphism — DNA fingerprinting	Individual variation in restriction sites → different fragment sizes → genetic profiling, paternity, forensics [PYQ 2011]

09 Antibiotic Targets

ANTIBIOTIC	TARGET	MECHANISM	SELECTIVE FOR
Rifampicin `Anticancer: none`	Prokaryotic RNA Pol (β subunit)	Binds β-subunit → blocks initiation of transcription (not elongation)	Prokaryotes; TB treatment
Actinomycin D `Anticancer`	DNA (intercalates between G-C)	Blocks movement of RNA Pol → inhibits transcription elongation in both pro and euk	Both (anticancer use)
α-Amanitin (Amanita phalloides)	RNA Pol II (eukaryotic)	Tightest binding → blocks translocation of Pol II. Also inhibits Pol III at higher doses; Pol I insensitive	Eukaryotes; toxin not antibiotic
Streptomycin	30S ribosome (16S rRNA / S12 protein)	Causes misreading of mRNA at A site → wrong aminoacyl-tRNA inserted → abnormal protein synthesis; also blocks initiation	Prokaryotes
Tetracycline	30S ribosome (A site)	Blocks binding of aminoacyl-tRNA·EF-Tu to A site → no new amino acid added	Prokaryotes (broad spectrum)
Chloramphenicol	50S ribosome (23S rRNA)	Inhibits peptidyl transferase activity → no peptide bond formation. Acts at A site	Prokaryotes; also mitochondria
Erythromycin	50S ribosome (23S rRNA / L4, L22)	Blocks translocation (EF-G step) → ribosome stalls	Prokaryotes
Linezolid	50S ribosome (23S rRNA)	Prevents formation of 70S initiation complex → blocks initiation	Prokaryotes (MRSA treatment)
Puromycin	Both 70S and 80S ribosomes	Structural analogue of aminoacyl-tRNA → incorporated at A site → premature chain termination (puromycyl-peptide released)	Both pro & euk → too toxic for clinical use; experimental tool
Ciprofloxacin	DNA Gyrase (Topoisomerase II) + Topoisomerase IV	Stabilises enzyme-DNA cleavage complex → double-strand breaks → bacterial death	Prokaryotes
Diphtheria toxin	EF-2 (eukaryotic EF-G equivalent)	ADP-ribosylates EF-2 (via NAD^+) → inactivates it → no translocation → blocks eukaryotic translation	Eukaryotes only [PYQ 2022]

🧠 30S INHIBITORS VS 50S INHIBITORS

"30S: STupid Tetracyclines" = Streptomycin, Tetracycline (+ spectinomycin, aminoglycosides)

"50S: CELls (50 = Fifty)" = Chloramphenicol, Erythromycin, Linezolid, cLindamycin

10 Post-translational Modifications

PTMs expand the proteome beyond what the genome encodes. They alter protein function, localisation, stability, and interactions.

MODIFICATION	ENZYME/PROCESS	EXAMPLE / SIGNIFICANCE
Removal of fMet / signal peptide	Methionine aminopeptidase / Signal peptidase	N-terminal processing; signal peptide directs protein to ER
Phosphorylation	Protein kinases (Ser, Thr, Tyr)	Most common reversible PTM; activates/deactivates enzymes (e.g., glycogen phosphorylase). Reversed by phosphatases
Glycosylation	Glycosyltransferases in ER + Golgi	N-linked (Asn) or O-linked (Ser/Thr); blood group antigens; protein folding (calnexin quality control)
Hydroxylation	Prolyl/lysyl hydroxylase (needs Vit C)	Collagen: hydroxyproline + hydroxylysine for stability and cross-linking. Scurvy = Vit C deficiency = defective collagen
Carboxylation	γ-carboxylase (Vit K-dependent)	Clotting factors II, VII, IX, X, Protein C, S → glutamate → γ-carboxyglutamate → binds Ca^{2+} + phospholipid
Acetylation	Acetyltransferases (histone HATs)	Histone acetylation → chromatin relaxation → ↑ transcription. Protein stability regulation
Ubiquitination	E1-E2-E3 ubiquitin ligase cascade	Tags proteins for proteasomal degradation; also involved in DNA repair, signalling
Proteolytic cleavage	Specific proteases	Insulin (preproinsulin → proinsulin → insulin + C-peptide); zymogen activation (trypsinogen → trypsin)
Disulphide bond formation	Protein disulphide isomerase (PDI) in ER	Stabilises protein 3° structure; IgG has interchain and intrachain S-S bonds
Myristoylation / Palmitoylation	N-myristoyltransferase	Lipid anchor → membrane association of signalling proteins (e.g., Src kinases)
Methylation	Methyltransferases	Histone methylation (gene regulation); DNA methylation (epigenetics, gene silencing)
ADP-ribosylation	Bacterial toxins / PARP	Cholera toxin (Gs protein), Diphtheria toxin (EF-2), Pertussis toxin (Gi)

COLLAGEN PTMS — VITAMIN C CONNECTION

Collagen is unusual: Gly-X-Y repeat. X = often Pro, Y = often Hyp (hydroxyproline). Proline hydroxylation requires Vitamin C (ascorbate) as cofactor for prolyl hydroxylase. Vit C deficiency → under-hydroxylated collagen → unstable triple helix → **Scurvy** (bleeding gums, poor wound healing, perifollicular haemorrhage).

11 Viva & Exam Q&A

REPLICATION & TRANSCRIPTION

Differentiate DNA Polymerase I, II and III in E. coli. `PYQ 2019, 2021`

Feature	DNA Pol I	DNA Pol II	DNA Pol III
Primary function	Primer removal + gap fill	DNA repair	Main replicase
Processivity	Low	Low	Very high (β-clamp)
5'→3' exonuclease	Yes (removes primers)	No	No
3'→5' exonuclease (proofreading)	Yes	Yes	Yes (ε subunit)
Speed	~20 nt/s	~5 nt/s	~1000 nt/s

Key exam point: ONLY Pol I has 5'→3' exonuclease for primer removal. Pol III is the workhorse. Pol II is the repair enzyme.

What is a promoter? What are the –10 and –35 boxes? `PYQ 2024`

A **promoter** is a DNA sequence upstream (5') of a gene where RNA Pol binds to initiate transcription.

Prokaryotic consensus sequences:

- **–35 box:** TTGACA (σ factor recognises this, initial binding)
- **–10 box (Pribnow box):** TATAAT (DNA melting — rich in A-T, 2 H-bonds, easy to melt)

Eukaryotic promoter elements (RNA Pol II):

- **TATA box** (~–25 bp): TATAAA; bound by TATA-Binding Protein (TBP, part of TFIID)
- **CAAT box** (~–75): GGCCAATCT; enhances basal transcription
- **GC box** (~–90): GGGCGG; bound by Sp1 transcription factor
- Enhancers: can be thousands of bp away, either orientation — increase transcription rate

LAC OPERON & REGULATION

Differentiate positive and negative regulation of the lac operon. PYQ 2015, 2023

Negative regulation (Repressor system):

- lac repressor (lacI gene product) binds operator → physically blocks RNA Pol → transcription OFF
- Allolactose (inducer) binds repressor → allosteric change → repressor leaves operator → RNA Pol can proceed
- "Negative" because the regulatory protein (repressor) inhibits transcription

Positive regulation (CAP-cAMP system):

- When glucose absent → adenylate cyclase active → cAMP rises
- cAMP binds CAP (catabolite activator protein) → conformational change → CAP-cAMP binds CAP site upstream of promoter
- CAP bends DNA ~90° → recruits RNA Pol → stabilises RNA Pol-promoter complex → transcription stimulated ~50×
- "Positive" because the regulatory complex activates transcription

Maximum transcription requires BOTH: allolactose removes repressor (negative control) AND glucose absence provides CAP-cAMP (positive control).

DNA REPAIR & MUTATIONS

What is Xeroderma Pigmentosum? What is its molecular basis? PYQ 2023

XP is an autosomal recessive disorder caused by **defective Nucleotide Excision Repair (NER)**.

Molecular basis:

- UV radiation (260–280 nm) → pyrimidine dimers (TT, TC dimers) in skin cell DNA
- Normal cells: XP proteins (XPA-XPG) recognise the helix distortion → recruit NER machinery → excise ~24–29 nt around dimer → fill gap → ligate
- XP patients: mutation in any XP gene → NER fails → dimers not repaired → accumulate with each UV exposure
- Unrepaired dimers → misreading during replication → C→T or CC→TT transitions (UV signature mutations)
- Mutations in tumour suppressors (p53) and proto-oncogenes → skin cancer

8 complementation groups (XPA–XPG = NER defects; XPV = Pol η deficiency = replication past unrepaired dimers → error-prone)

What is the difference between a transition and a transversion mutation? PYQ 2014-S

Transition: Substitution within the same chemical class of base.

- Purine → Purine: A↔G
- Pyrimidine → Pyrimidine: C↔T (or C↔U in RNA)
- More common (2/3 of point mutations)
- Example: Deamination of 5-methylcytosine → Thymine = C→T transition; most common spontaneous mutation

Transversion: Substitution between different chemical classes.

- Purine → Pyrimidine or vice versa: A↔C, A↔T, G↔C, G↔T
- Less common (1/3 of point mutations)
- Example: HbS mutation — A→T transversion at codon 6 of β-globin

What is p53? Why is it called the "guardian of the genome"? PYQ 2019-S, 2025

p53 is a tumour suppressor protein (encoded by TP53 gene on chromosome 17p13.1).

Why "guardian of genome":

- Normally kept at low levels by MDM2-mediated ubiquitination → proteasomal degradation
- DNA damage or oncogenic stress → ATM/ATR kinases phosphorylate p53 → MDM2 cannot bind → p53 stabilised + activated
- **G1/S checkpoint:** p53 transcribes p21 (CDK inhibitor) → CDK2-CyclinE blocked → Rb stays unphosphorylated → E2F transcription factor not released → cell cannot enter S phase → time for repair
- **Apoptosis:** If damage irreparable → p53 transcribes BAX, PUMA, NOXA → mitochondrial apoptosis pathway
- **Senescence:** Chronic activation → permanent cell cycle arrest

p53 is mutated in ~50% of all human cancers — most frequently mutated tumour suppressor gene. Li-Fraumeni syndrome = germline p53 mutation → diverse cancers at young age.

12 Essay Question Templates

EQ — Genetic Code is Degenerate [PYQ 2012-S, 2018]
"Genetic code is degenerate."

1. **Definition:** Degeneracy (redundancy) means that most amino acids are encoded by more than one codon. There are 64 codons (4^3) for only 20 amino acids + 3 stop codons.
2. **Distribution:** 2 amino acids have only 1 codon: Methionine (AUG — also start) and Tryptophan (UGG). Most others have 2–6 codons (e.g., Leu, Ser, Arg each have 6 codons).
3. **Molecular basis — Wobble hypothesis (Crick):** The 3rd position (3' end) of the codon is the "wobble position." Unconventional base pairing is tolerated here. A single tRNA can recognise multiple codons differing at the 3rd position. Example: inosine at wobble position of anticodon can pair with U, C, or A.
4. **Why degeneracy is beneficial:** Provides protection against point mutations — many base changes at 3rd codon position → same amino acid (silent/synonymous mutation) → protein unchanged. This is "mutational buffering."
5. **Exceptions — non-degenerate:** Met and Trp each have only 1 codon → mutations here always result in amino acid change.
6. **Important distinction:** Degeneracy ≠ ambiguity. Code is non-ambiguous — one codon specifies only ONE amino acid (no overlap in the other direction).

EQ — PCR: Steps, Reagents, Applications [PYQ 2011-S, 2021]

"Describe the principles of PCR. Enumerate reagents and applications."

1. **Principle:** In vitro amplification of a specific DNA sequence using thermostable DNA polymerase and two oligonucleotide primers. Based on repeated thermal cycling — each cycle doubles the target → exponential amplification (2^n copies after n cycles).
2. **Reagents:** Template DNA; forward primer (complementary to one strand); reverse primer (complementary to other strand); Taq polymerase (Thermus aquaticus, stable at 95°C); dNTPs; $MgCl_2$ (cofactor); buffer; thermocycler.
3. **3 Steps per cycle:** Denaturation (94–96°C, ~30 sec → strands separate); Annealing (50–65°C, ~30 sec → primers bind); Extension (72°C, ~1 min/kb → Taq synthesises new strand 5'→3').
4. **Product accumulation:** First few cycles produce long "background" products. By cycle 3, short "specific" products bounded by primers accumulate exponentially. Typical 30–35 cycles → ~10^8-fold amplification.
5. **Applications:** (a) Diagnosis: TB (gold standard), HIV, hepatitis viruses, COVID-19; (b) Forensics: DNA fingerprinting using STRs; (c) Prenatal diagnosis of genetic diseases; (d) Detection of mutations (BRCA1/2, KRAS); (e) Research: cloning, gene expression (RT-PCR); (f) Blood banking: donor screening; (g) Epidemiology: pathogen genotyping.
6. **Limitation:** Contamination of even 1 molecule can give false positive; cannot detect unknown sequences without prior sequence information.

EQ — Sickle Cell Anaemia is a Molecular Disease [PYQ 2023]

"Sickle cell anaemia is a molecular disease."

1. **Historical context:** First described as a "molecular disease" by Linus Pauling (1949) using electrophoresis — HbS moves differently from HbA. Mechanism elucidated by Vernon Ingram (1956) — tryptic peptide mapping revealed single amino acid difference.
2. **Genetic defect:** Point mutation in β-globin gene (chromosome 11) — single A→T transversion at codon 6 — GAG (Glu) → GTG (Val). This is a missense mutation causing an amino acid substitution.
3. **Structural consequence:** Glutamate (charged, hydrophilic) → Valine (non-polar, hydrophobic) at position 6 of β-chain. Creates a hydrophobic "sticky patch" on the surface of deoxyHbS.
4. **Molecular pathology:** Deoxygenation → HbS in T-state exposes hydrophobic patch → fits into complementary pocket on adjacent deoxyHbS molecule → polymerisation → long fibrous tactoids → RBC distorts into sickle shape.
5. **Clinical consequences:** Haemolytic anaemia (sickled cells destroyed in spleen, RBC lifespan 20 days vs 120 days); vaso-occlusion (rigid sickled cells block microvasculature → pain crises, organ infarction); splenic sequestration and autosplenectomy.
6. **Why "molecular disease":** A single nucleotide change in the genome → single amino acid substitution in the protein → altered protein behaviour → systemic disease. Exemplifies the gene → protein → disease paradigm.

🔥 EXAM DAY HIGH-YIELD QUICKFIRE — MODULE 06

QUESTION	ANSWER
Main replicase in E. coli	DNA Polymerase III (α-subunit synthesises, ε proofreads)
Primer removal in prokaryotes	DNA Pol I (5'→3' exonuclease activity)
Telomerase substrate	RNA-dependent DNA polymerase (reverse transcriptase) — extends chromosome ends in cancer cells + stem cells
Only RNA Pol in prokaryotes	One (core $\alpha_2\beta\beta'\omega$ + σ = holoenzyme)
α-Amanitin inhibits	RNA Pol II (most sensitive); Pol III (high dose); Pol I (insensitive)
Shine-Dalgarno sequence	Prokaryotic mRNA sequence that binds 16S rRNA of 30S subunit — positions ribosome at AUG
Peptidyl transferase is a...	Ribozyme (23S rRNA in prokaryotes, 28S rRNA in eukaryotes)
Diphtheria toxin target	ADP-ribosylates EF-2 → blocks eukaryotic translation
Rifampicin mechanism	Inhibits β-subunit of prokaryotic RNA Pol → blocks initiation of transcription
p53 checkpoint control	Induces p21 (CDK inhibitor) → G1/S arrest for DNA repair; if irreparable → BAX/PUMA → apoptosis
Lac operon: max transcription requires	Lactose present (removes repressor) + Glucose absent (cAMP↑ → CAP activated)
NER disease	Xeroderma Pigmentosum (XPA-XPG); pyrimidine dimers unrepaired → skin cancer
MMR disease	Lynch syndrome (HNPCC) — MSH2, MLH1 mutations → hereditary colorectal cancer
Puromycin specificity	Inhibits BOTH 70S and 80S → too toxic for therapy; experimental only
Southern → Northern → Western	DNA → RNA → Protein detection (gene → transcript → protein)

MODULE 06 MASTER MNEMONICS

RIBOSOME SITES

"APE" = Aminoacyl → Peptidyl → Exit

STOP CODONS

"UAA, UAG, UGA — U Are Away, U Are Gone, U Go Away"

🧠 30S VS 50S ANTIBIOTICS

"STupid Tetracyclines" at 30S; "CELLs" (Chloramphenicol, Erythromycin, Linezolid, cLindamycin) at 50S

🧠 BLOTTING METHODS

"SNoW DRoP" = Southern(DNA), Northern(RNA), Western(Protein)

MODULE 06 · MOLECULAR BIOLOGY & GENETICS

Caffeine & Cadaver · MBBS Biochemistry Notes · 2010-2025

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